

MAHA VAARSHINIE ARUPATHA RAJOO

Data Scientist | AI/ML Engineer | Data Analyst | Software Engineer

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PROFESSIONAL SUMMARY

Computer Science (Data Science) graduate from Multimedia University who builds end to end data products across ETL pipelines, machine learning models, and business intelligence dashboards. Final Year Project on deepfake detection ranked 5th of 279 projects, won an international Silver Medal at iNVENTX, and was presented at DIFCON 2026. Built operations wide Power BI reporting across every department of a multinational high tech manufacturer at AT&S, covering production, shipping, and workforce metrics, with Cleanroom Protocol and Facility Management Control training alongside it. Works comfortably across the full span of a data role, from SQL and ETL through Python modelling in PyTorch and Hugging Face to dashboard delivery in Power BI, Tableau, and D3.js. Semifinalist in two international hackathons and an active student leader on university committees since Foundation year. Seeking Data Science, AI/ML Engineering, Data Analyst, or Software Engineering roles.

TECHNICAL SKILLS

Data Visualization and BI: Power BI (DAX, Power Query), Tableau, D3.js, Advanced Excel (VBA Macros, Pivot Tables), Matplotlib, Plotly, Dashboard Design, Data Storytelling

Data Analytics and Engineering: SQL, ETL Pipelines, Data Modelling (Star Schema), Data Warehousing, Advanced Database Management, Statistical Analysis, Pandas, NumPy, PostgreSQL

Machine Learning and AI: PyTorch, scikit-learn, Deep Learning, Computer Vision, NLP, CNN, LSTM, GAN, Transformers (BERT), Hugging Face, spaCy, OpenCV, Optuna Tuning, Model Deployment (ONNX Runtime)

Agentic AI and LLM: LangChain, Model Context Protocol (MCP), Agentic AI Workflows, Prompt Engineering, n8n Automation

Software Engineering: Python, Java, C++, JavaScript, TypeScript, FastAPI, Flask, Django, React.js, Next.js, REST APIs, Git, Docker, Agile

Manufacturing Domain: Cleanroom Protocol, Facility Management (FM) Control, Process and Cycle Time Analytics, Yield and Quality Reporting, Process Automation

PROFESSIONAL EXPERIENCE

AT&S Austria Technologie & Systemtechnik (Malaysia)

IT Intern, Operations

Kulim, Malaysia

Jul 2025 - Oct 2025

- Built Power BI dashboards with DAX and Power Query for every department under Operations, including Test, Assembly, and SRSF, tracking manufacturing output, shipping, process time, waiting time, and target versus actual performance.
- Developed an employee attendance dashboard from clock in and clock out records and automated recurring reports with Excel VBA macros and reusable templates.
- Managed and queried SQL databases for operations reporting. Completed internal modules in Cleanroom Protocol and Facility Management (FM) Control.

EDUCATION

Multimedia University (MMU)

Bachelor of Computer Science (Honours), Data Science | CGPA 3.59

Cyberjaya, Malaysia

Aug 2023 - Jul 2026

- Final Year Project ranked 5th of 279 projects and awarded Best Project, Data Science category.
- Assistant Head, Residents' Welfare Division.** Students' College Committee. Led welfare initiatives and residential operations, staying active across student bodies and campus events.
- Host and Trainer, Power BI Workshop.** DATAI Community. Designed and taught a hands on session covering dashboard design, DAX, and data storytelling.

Multimedia University (MMU)

Foundation in Information Technology | CGPA 3.52

Cyberjaya, Malaysia

Aug 2022 - Jul 2023

- Committee Member, Business Operations.** Students' College Committee. Planned and ran college business operations and campus events.

AWARDS AND ACHIEVEMENTS

- Silver Medal (International),** iNVENTX International Invention and Innovation Competition.
- Best Project Award,** Data Science category, ranked 5th of 279 Final Year Projects, Multimedia University.
- Semifinalist, International Hackathons,** VHack and UMHackathon.

PROJECTS

Deepfake Detection Using Machine Learning Models | Final Year Project

- Dual stream CNN pairing ResNet-50 and Xception backbones to detect spatial artifacts and micro texture inconsistencies, reaching 98.05% accuracy on FaceForensics++ and beating single stream baselines.

Multi-Level Sentiment Analysis and Aspect-Based Opinion Mining | Big Data NLP

- Co-developed an ETL and NLP pipeline over 200,000+ Flipkart reviews benchmarking nine models, reaching 89.25% accuracy and 76.10% macro F1 with an Optuna tuned BERT, plus a spaCy ABSA module for aspect level drivers.

SmartStock AI | Inventory Decision Support System

- FastAPI and React.js decision support system forecasting SME demand from historical sales and converting it into reorder quantities, timing, and stock out risk scores.

The Inclusive Citizen | Multilingual AI for Public Services

- FastAPI NLP chatbot on the SEA-LION dataset answering public service queries for under represented ASEAN communities in low resource regional dialects.

PUBLICATIONS

"Deepfake Detection Using Machine Learning Models," presented at the Digital Futures International Congress (DIFCON) 2026. In press, AIP Conference Proceedings.

CERTIFICATIONS

- Microsoft Power BI Data Analyst Associate
- Microsoft Azure AI Essentials (Microsoft and LinkedIn)
- Huawei HCIA-Cloud Computing V5.5
- AWS Educate Machine Learning Foundations
- Advanced NLP with Python for Machine Learning
- IBM MQ Developer Essentials